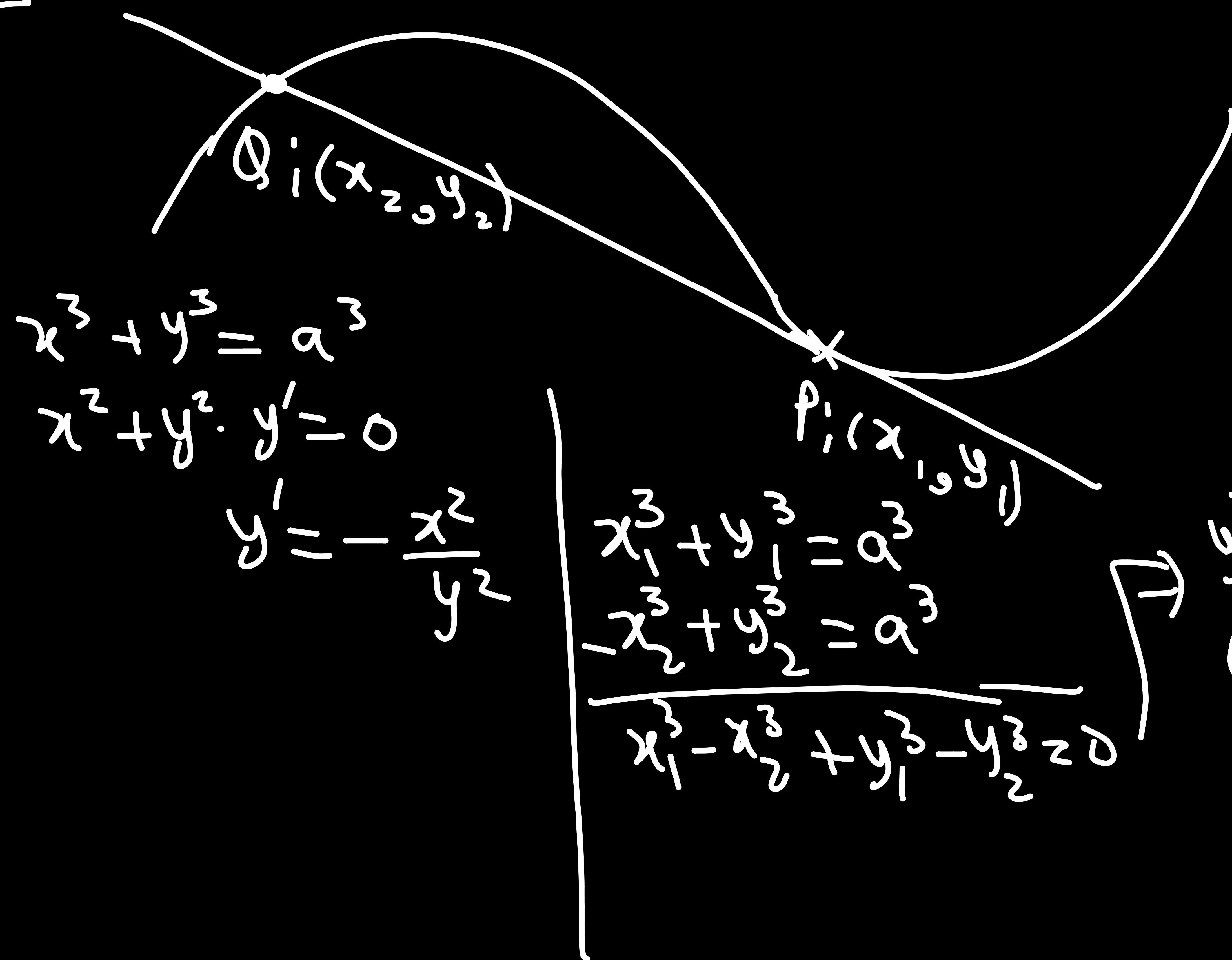


Que If the tangent at (x_1, y_1) to the curve $x^3 + y^3 = a^3$ meets the curve again at (x_2, y_2) then find $\frac{x_2}{x_1} + \frac{y_2}{y_1}$.

Sol



$$\Rightarrow y_2^3 - y_1^3 = -(x_2^3 - x_1^3)$$

$$\frac{y_2 - y_1}{x_2 - x_1} = -\frac{x_1^2 + x_2^2 + x_1 x_2}{y_1^2 + y_2^2 + y_1 y_2}$$

$$m_{PQ} = m_{PQ}$$

$$-\frac{x_1^2}{y_1^2} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$-\frac{x_1^2}{y_1^2} = -\frac{x_1^2 + x_2^2 + x_1 x_2}{y_1^2 + y_2^2 + y_1 y_2}$$

$$1 = \frac{1 + \frac{x_2^2}{x_1^2} + \frac{x_2}{x_1}}{1 + \frac{y_2^2}{y_1^2} + \frac{y_2}{y_1}}$$

$$\downarrow \frac{x_2}{x_1} = \alpha$$

$$\textcircled{Q} \frac{y_2}{y_1} = \beta$$

$$\frac{1+\alpha^2+\alpha}{1+\beta^2+\beta} = 1 \Rightarrow 1+\alpha^2+\alpha = 1+\beta^2+\beta$$

$$\alpha^2 - \beta^2 + \alpha - \beta = 0$$

$$(\alpha - \beta)(\alpha + \beta + 1) = 0$$

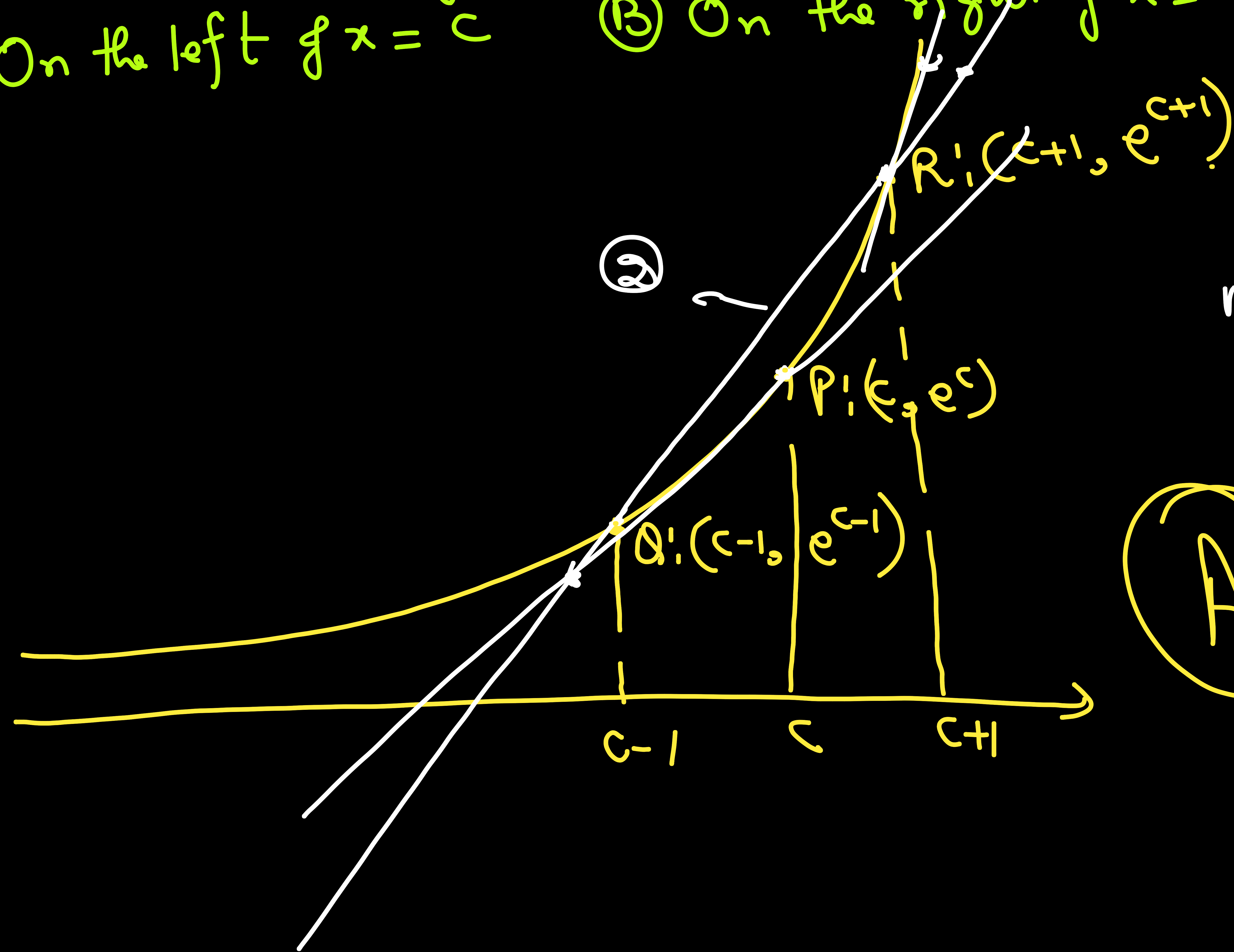
$$\alpha \neq \beta \Downarrow$$

$$\alpha + \beta + 1 = 0$$

$$\frac{x_2}{x_1} + \frac{y_2}{y_1} = \boxed{-1} \quad \underline{A}$$

Ques The tangent to the curve $y = e^x$ drawn at (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$,
 (A) On the left of $x = c$ (B) On the right of $x = c$ (C) at no point (D) both sides.

Sol



$$m_t^P = (e^x)'_{x=c} = e^c.$$

$$m_{RQ} = \frac{y_R - y_Q}{x_R - x_Q} = \frac{e^{c+1} - e^{c-1}}{c+1 - (c-1)}$$

$$= e^c \left(\frac{e - e^{-1}}{2} \right) > e^c = m_t^P.$$

(A)

Ques Find the interval of increase of $g(x)$ if $g(x) = f(x^2 - 2x + 8) + f(14 + 2x - x^2)$
where $f''(x) \geq 0, \forall x \in \mathbb{R}$.

Soln $f''(x) \geq 0 \Rightarrow f' \uparrow$

$$g(x) = f(x^2 - 2x + 8) + f(14 + 2x - x^2)$$
$$g'(x) = (2x - 2) \cdot f'(x^2 - 2x + 8) - (2x - 2) \cdot f'(14 + 2x - x^2)$$

$$= 2(x-1) \cdot \underbrace{f'(x^2 - 2x + 8) - f'(14 + 2x - x^2)}_{= \alpha}$$

if $\alpha > 0$ then

$$\begin{cases} f'(x^2 - 2x + 8) > f'(14 + 2x - x^2) \\ \Rightarrow x^2 - 2x + 8 > 14 + 2x - x^2 \Rightarrow x^2 - 2x - 3 > 0 \\ \quad \quad \quad (x+1)(x-3) > 0 \end{cases}$$
$$x \in (-\infty, -1) \cup (3, \infty)$$

$$\alpha > 0 \Rightarrow x \in (-\infty, -1) \cup (3, \infty)$$

$$\alpha < 0 \Rightarrow x \in (-1, 3)$$

Sign of $g'(x)$

$\alpha \rightarrow$	+		-		+		+
$\beta \rightarrow$	-		-		+		+
$g' \rightarrow$	-		+		-		+
		-1		1		3	

So $g(x)$ increases in $[-1, 1]$ and in $[3, \infty)$.

Ans

Ques Find 'K' if the equation $x^3 - 3x + K = 0$ has two distinct roots in $(0, 1)$.

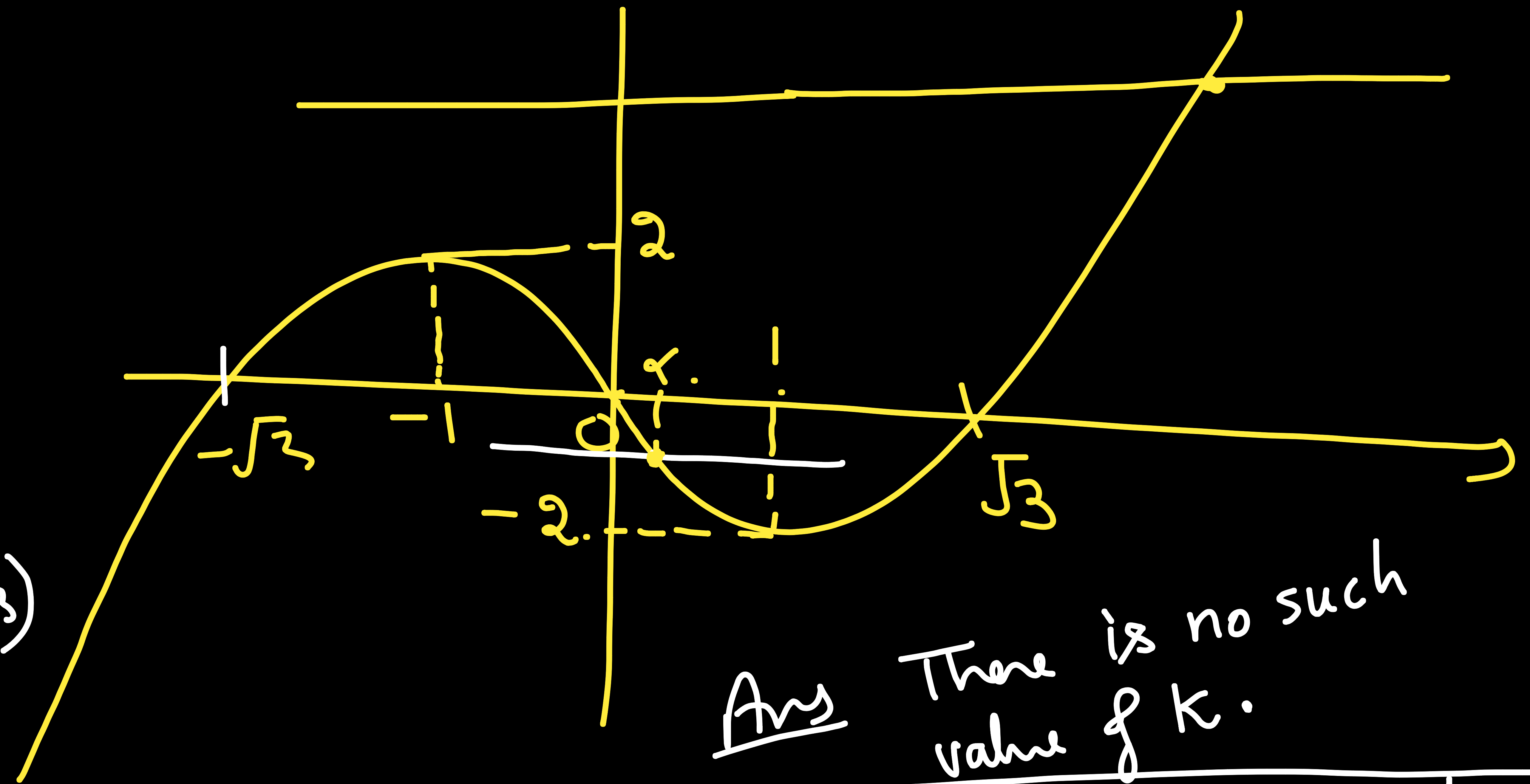
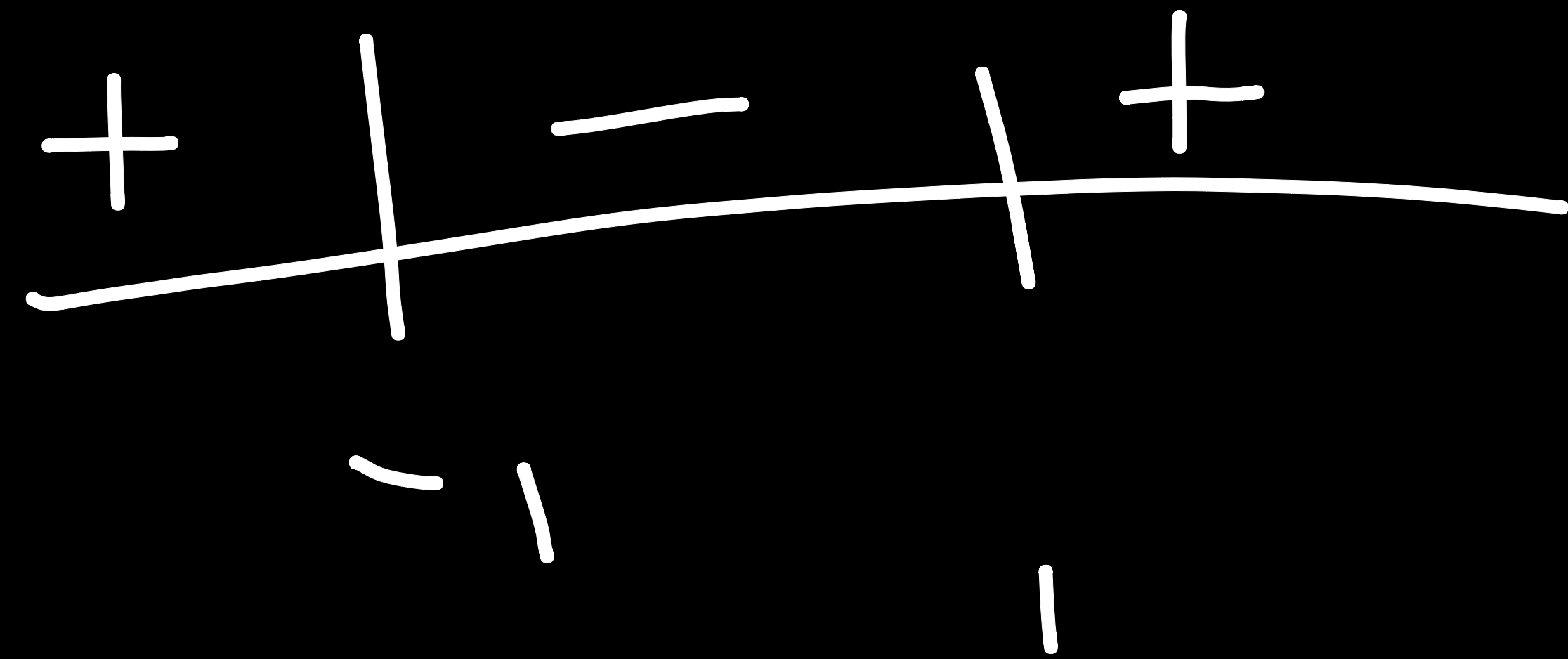
Sol

$$x^3 - 3x + K = 0$$

$$\underbrace{x^3 - 3x}_y = \underbrace{-K}_y$$

$$y = x^3 - 3x = x(x - \sqrt{3})(x + \sqrt{3})$$

$$y' = 3x^2 - 3 = 3(x - 1)(x + 1)$$



Ans There is no such value of K.

S.O. find 'K' if eqn. has atleast one root in $(0, 1)$

$$-K \in (-2, 0) \Rightarrow \boxed{K \in (0, 2)}$$

Q

Q

Ques Find the largest term of the sequence $a_n = \frac{n^2}{n^3 + 200}$, where $n \in \mathbb{N}$.

Sol

$$a_n = \frac{n^2}{n^3 + 200}$$

Let $f(x) = \frac{x^2}{x^3 + 200}$

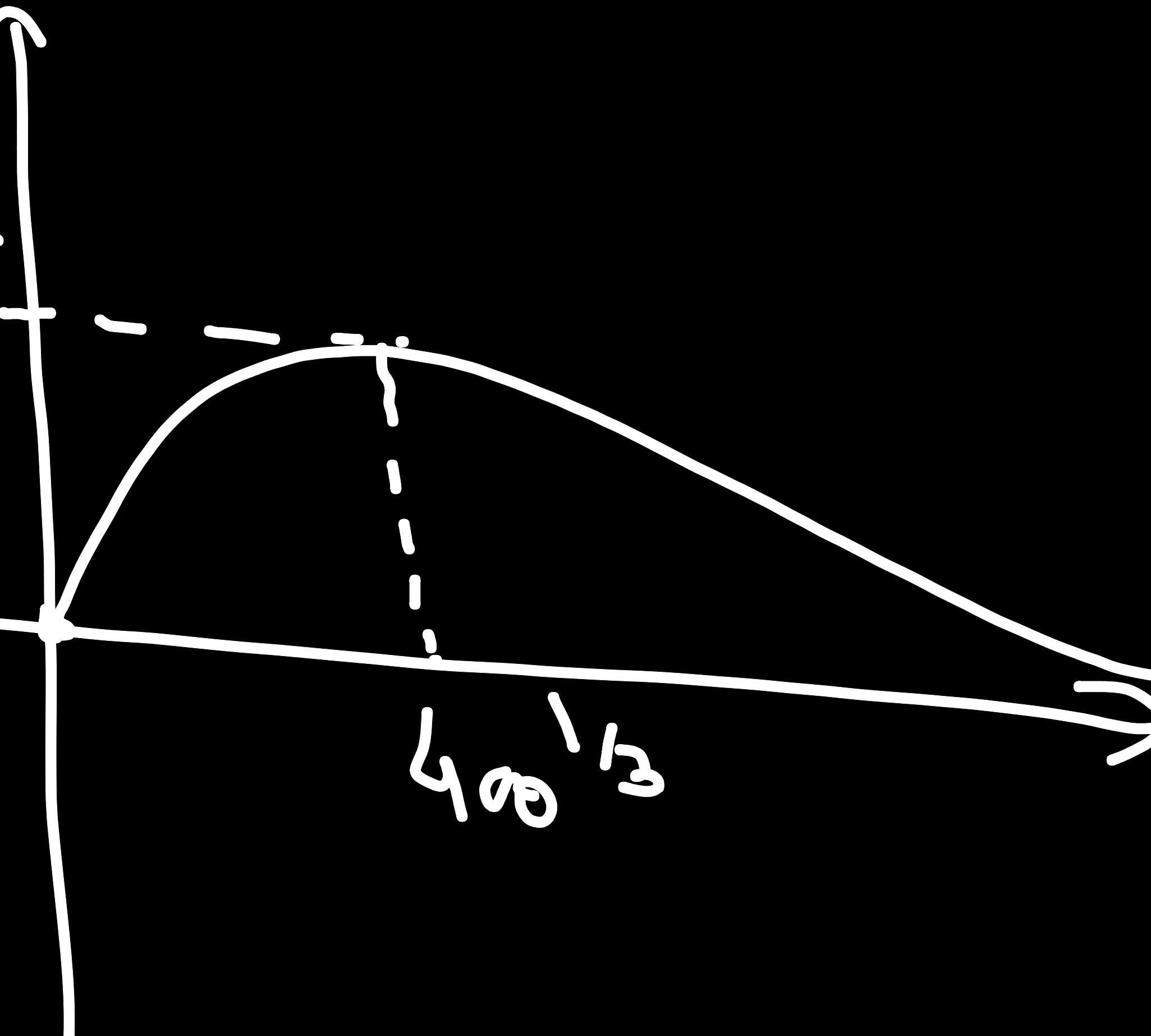
$$f'(x) = \frac{2x(x^3 + 200) - 3x^2 \cdot x^2}{(x^3 + 200)^2}$$

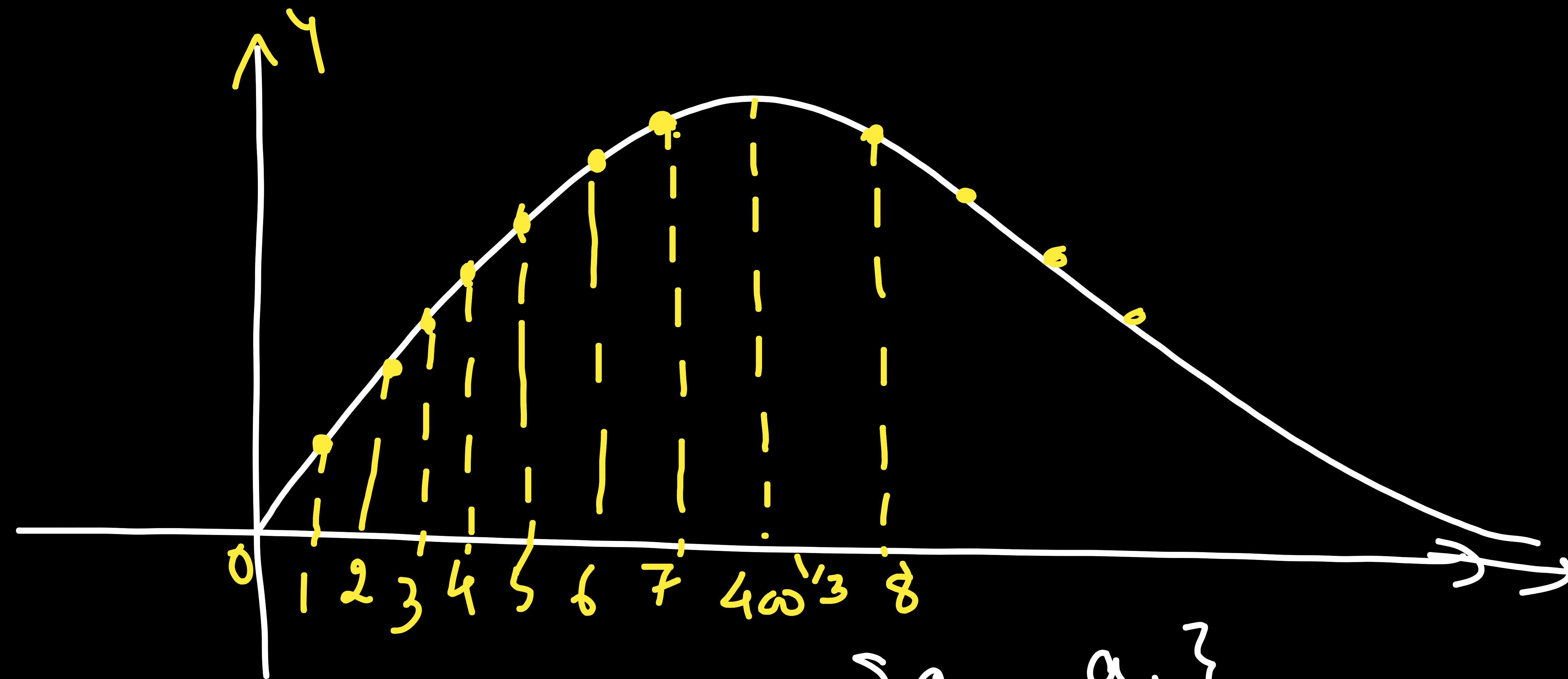
$$= 0$$

$$x(2x^3 + 400 - 3x^3) = 0$$

$$= 0$$

$$\Rightarrow 400 = x^3$$

$$x = (400)^{1/3}$$




$$(a_n)_{\max.} = \max. \{a_7, a_8\}$$

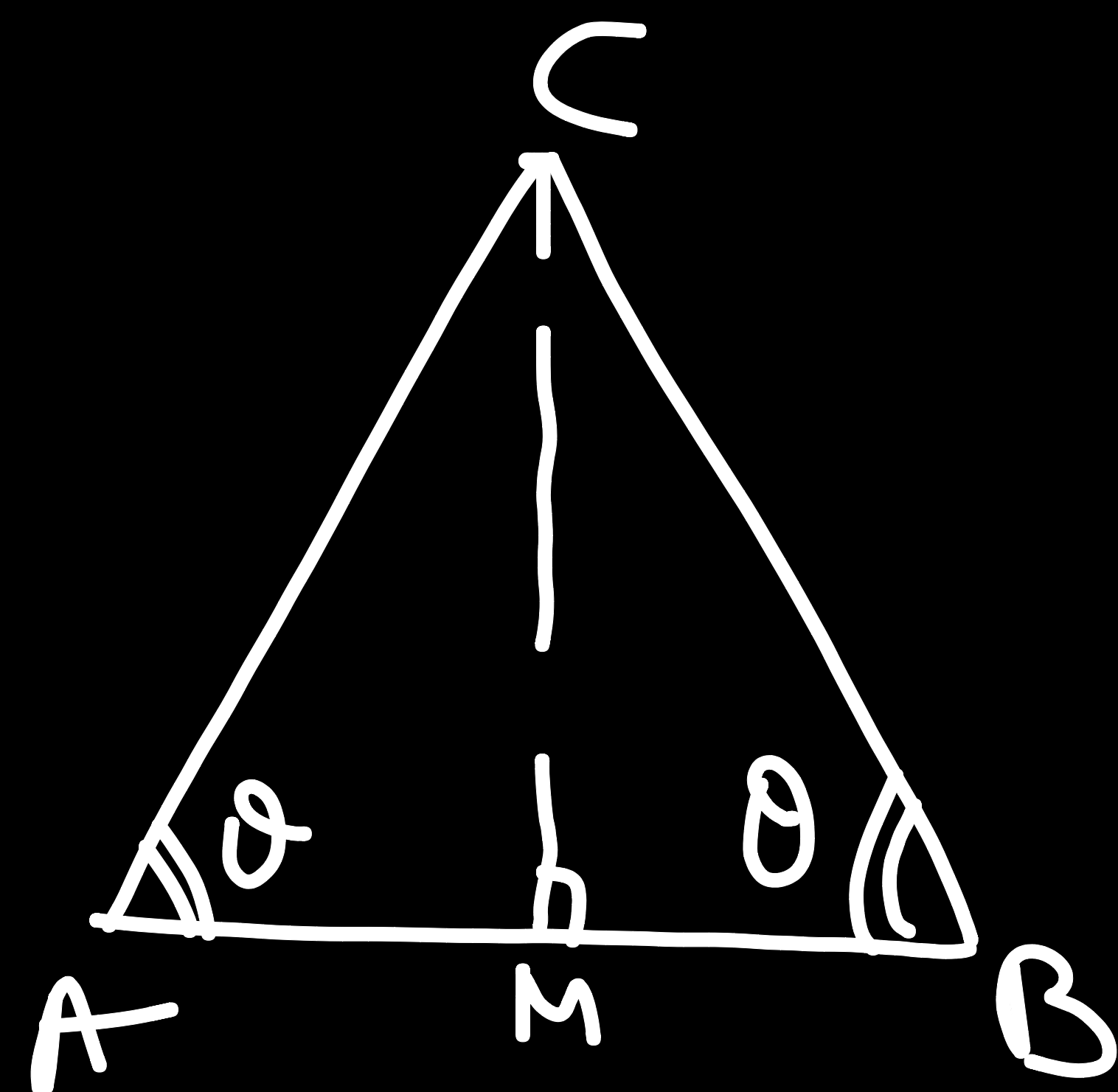
$$a_7 = \frac{49}{343 + 200} = \frac{49}{543} \approx \frac{1}{11}$$

$$a_8 = \frac{16}{512 + 200} = \frac{16}{712} \approx \frac{1}{44.5}$$

$$(a_n)_{\max.} = \frac{49}{543}$$

Ques Find number of non-similar isosceles triangles ABC , such that $\tan(A) + \tan(B) + \tan(C) = 100$.

Sol



Let $A = B = \theta$
 $\theta \in (0, 90^\circ)$

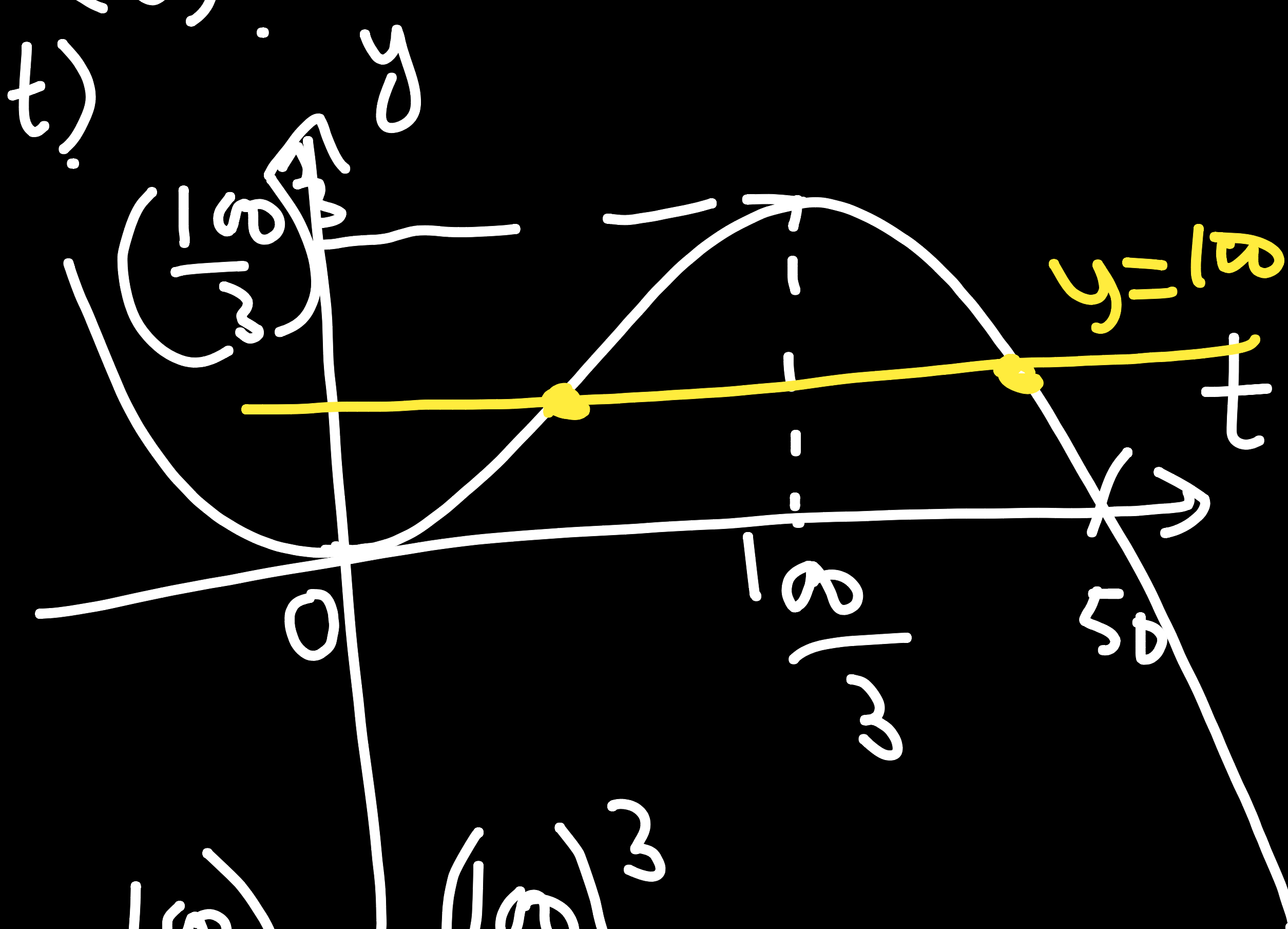
$\tan(\theta) = \tan(A) = \tan(B) = t$
 $t \in (0, \infty)$ (say)

$\Rightarrow t + t + \tan(C) = 100 \Rightarrow \tan(C) = 100 - 2t.$

Also in a Δ
 $\sum \tan(A) = n \tan(A)$

$\Rightarrow 100 = t \cdot t \cdot (100 - 2t)$
 $100 = 2t^2(50 - t)$

$y' = 2(2t(50 - t) - t^2)$
 $= 2t(100 - 3t)$

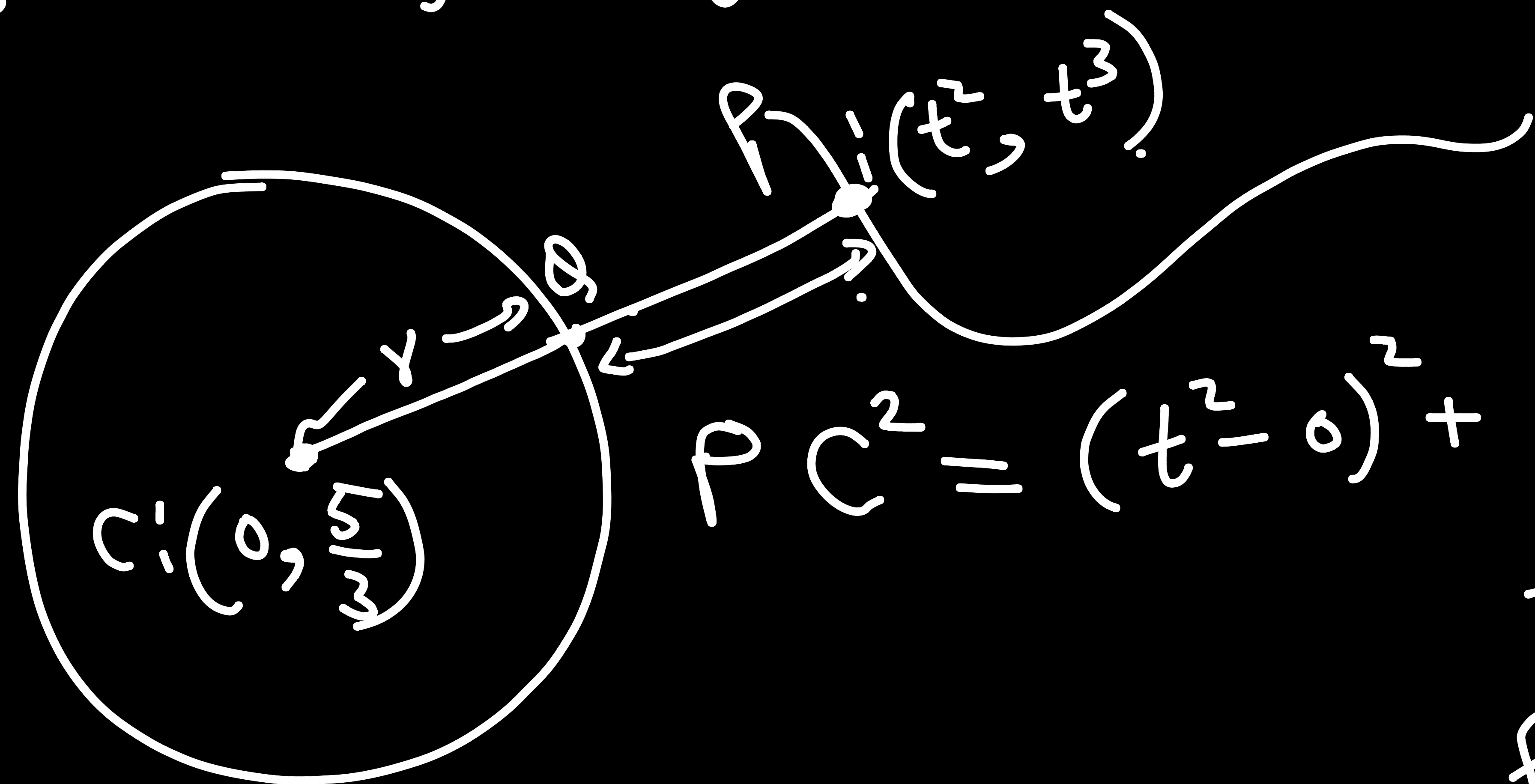


Required D's $= 2$

Ques Find the shortest distance between $9x^2 + 9y^2 - 30y + 16 = 0$ and $y^2 = x^3$.

Sol

$$9x^2 + 9y^2 - 30y + 16 = 0 \quad \textcircled{1} \quad y^2 = x^3 \quad \textcircled{2}$$



$$r = \sqrt{\left(\frac{5}{3}\right)^2 - \frac{16}{9}}$$

$$= \sqrt{\frac{25-16}{9}} = 1.$$

$$PC^2 = (t^2 - 0)^2 + \left(t^3 - \frac{5}{3}\right)^2 = t^4 + t^6 + \frac{25}{9} - \frac{10}{3}t^3 = f(t) \text{ (say)}$$

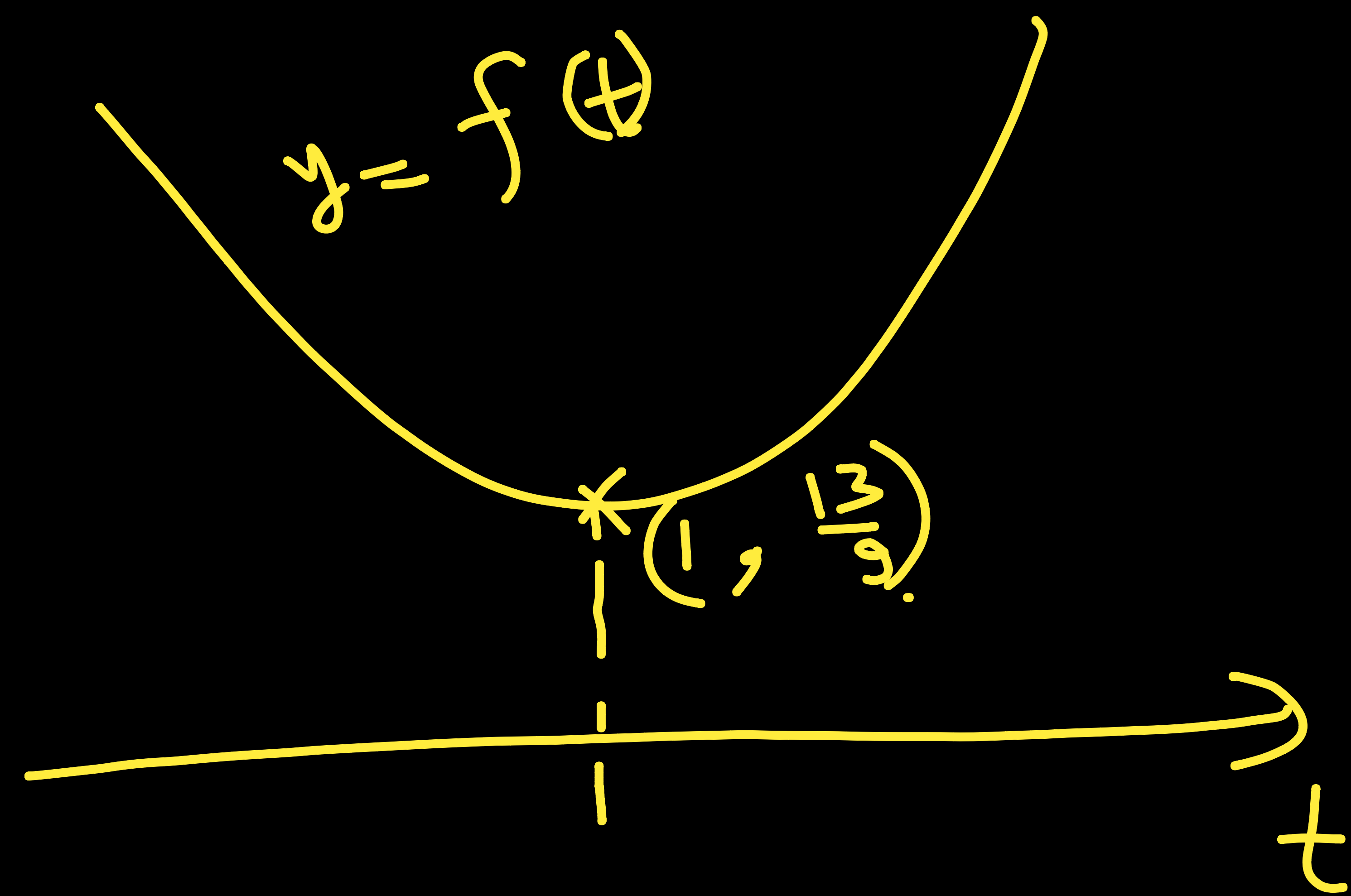
$$f(t) = t^6 + t^4 - \frac{10}{3}t^3 + \frac{25}{9}$$

$$f'(t) = 6t^5 + 4t^3 - 10t^2 = t^2(6t^3 + 4t - 10)$$

$$= t^2(t-1)(6t^2 + 6t + 10) = 2t^2(t-1)(3t^2 + 3t + 5)$$

$t=1$ is a local min.

$$f(1) = 1 + 1 - \frac{10}{3} + \frac{25}{9} = \frac{18 - 30 + 25}{9} = \frac{13}{9}$$



Required min. distance = $PQ = PC - r = \frac{\sqrt{13}}{3} - 1 = \boxed{\frac{\sqrt{13} - 3}{3}}$

A

Ques Find the minimum and maximum value of $f(x, y)$ where
 (i) $f(x, y) = (x - y)^2 + (x^2 - 4y - 7)^2$ (ii) $f(x, y) = (x - y)^2 + (x^2 - 4y + 7)^2$
 $x, y \in \mathbb{R}$.

Sol (i) $f(x, y) = (x - y)^2 + (x^2 - 4y - 7)^2$.
 If $(x - y) = 0$ ① $x^2 - 4y - 7 = 0 \Rightarrow x^2 - 4x - 7 = 0$ roots x_1, x_2 .
 $\hookrightarrow x = y$ $\hookrightarrow D > 0 \Rightarrow \underline{x \in \mathbb{R}}$.

$\Rightarrow [f(x, y)]_{\min} = 0$ for $x = y = x_1, x_2$
 $(f(x, y))_{\max} = \text{D.N.E. } (\rightarrow \infty)$.

(ii) $f(x, y) = (x - y)^2 + (x^2 - 4y + 7)^2$

Check $x = y$ ② $x^2 - 4x + 7 = 0$
 $\hookrightarrow D < 0 \Rightarrow \text{No real root.}$

$\Rightarrow f(x, y) = 0$ not possible.

$$f(\alpha, \beta) = (\alpha - \beta)^2 + (\alpha^2 - 4\beta + 7)^2$$

$$= (x_1 - x_2)^2 + (y_1 - y_2)^2$$

Assume

$$P: (x_1, y_1) \equiv (\underset{\text{"x"}}{\alpha}, \underset{\text{"y"}}{\alpha^2}) \longrightarrow y = x^2$$

$$Q: (x_2, y_2) \equiv (\underset{\text{"x"}}{\beta}, \underset{\text{"y"}}{4\beta - 7}) \longrightarrow y = 4x - 7$$

$$4x - y - 7 = 0$$

$$\Rightarrow f(\alpha, \beta) = PQ^2$$

for A

$$y = x^2 \xrightarrow{\frac{d}{dx}} y' = 2x = 4$$

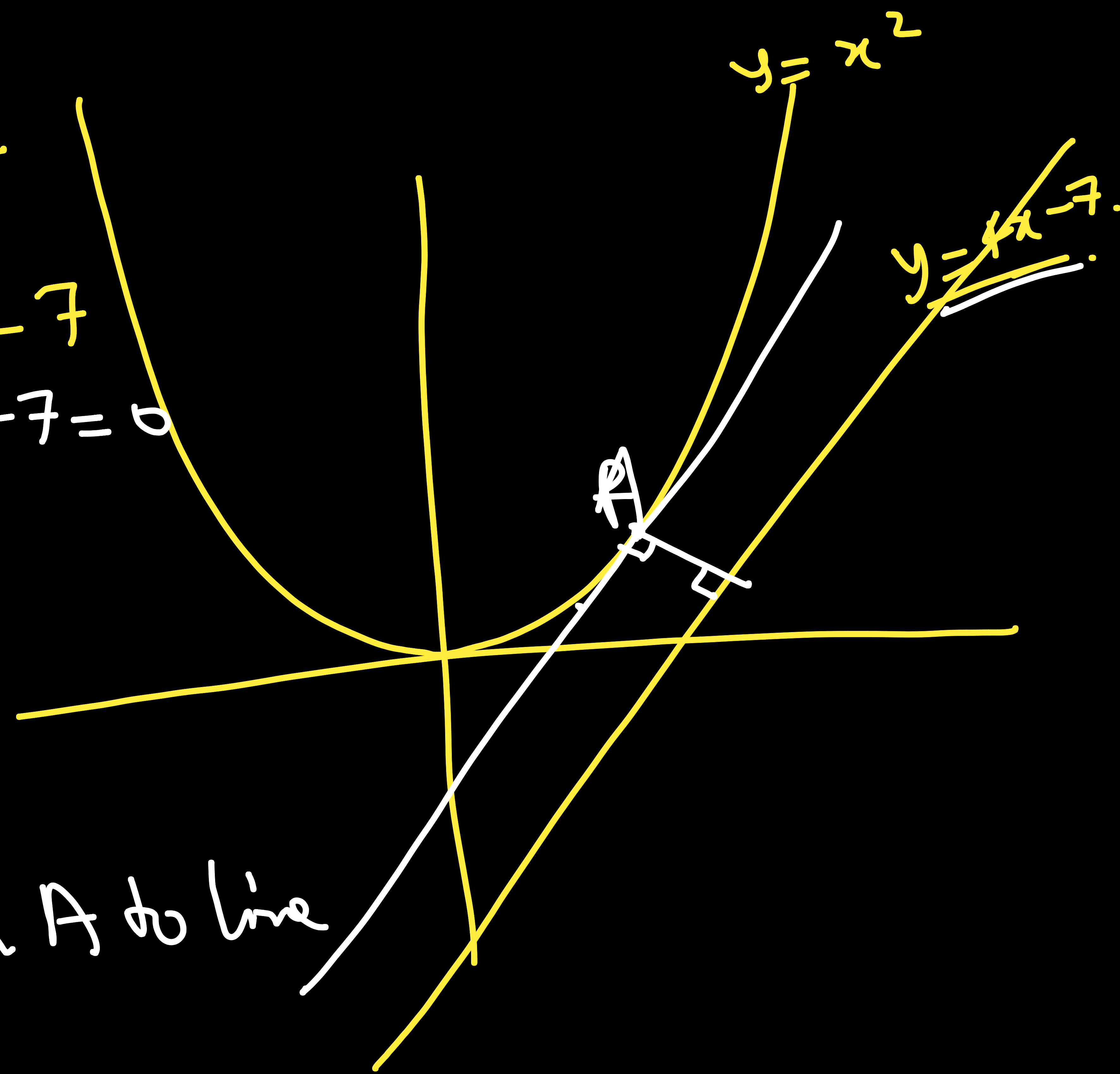
$$\Rightarrow x = 2$$

$$A: (2, 4)$$

Minimum distance $PQ =$ length of $\perp$ from A to line

$$\Rightarrow f(\alpha, \beta)_{\min} = PQ_{\min}^2 = \left(\frac{9}{17} \right) \frac{1}{PQ} = \frac{3}{\sqrt{17}}$$

$$f(\alpha, \beta)_{\max} = \text{D.N.E.} \quad (\rightarrow \infty \uparrow)$$



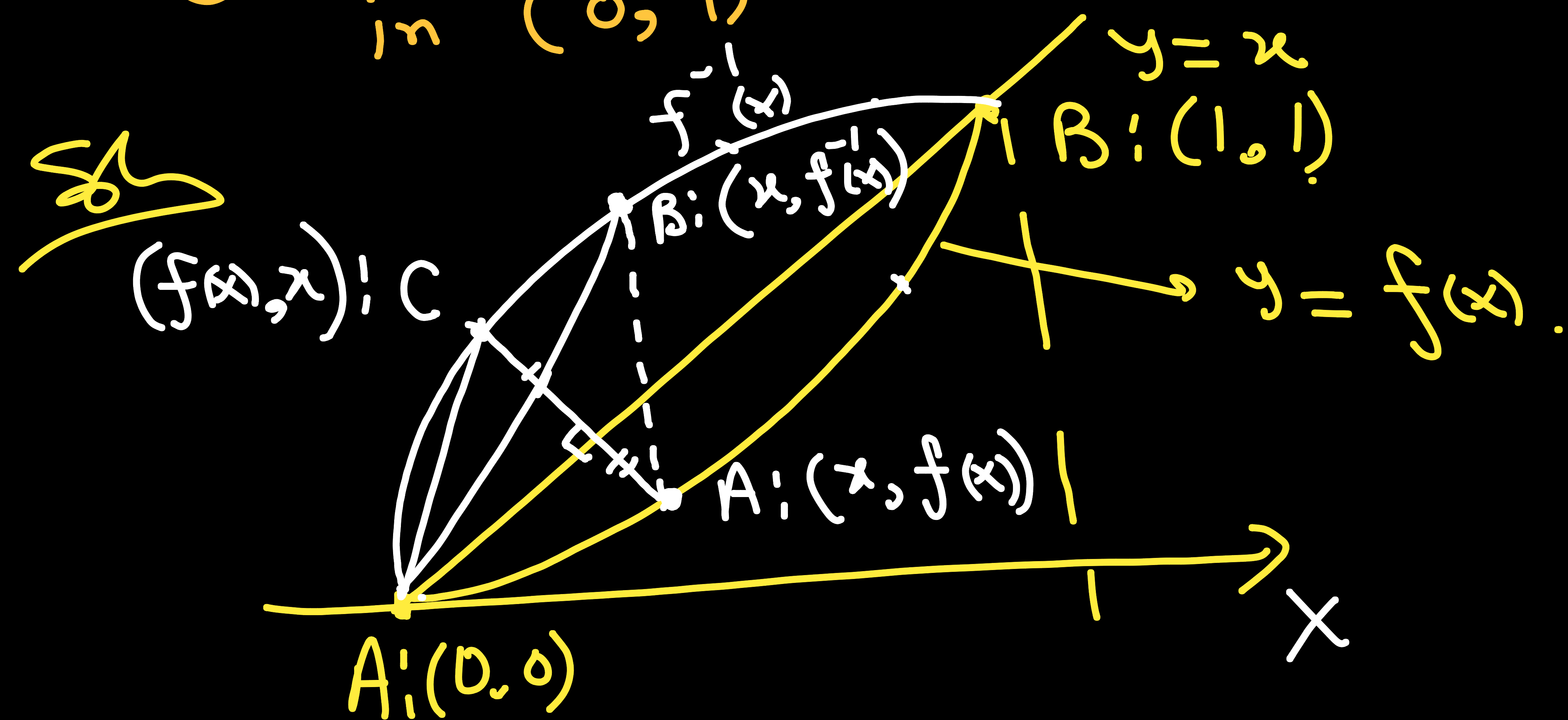
Ques If $f'(x) > 0$, $f''(x) > 0$, $\forall x \in (0, 1)$ and $f(0) = 0$ and $f(1) = 1$ then

(A) $f(x) \cdot f^{-1}(x) < x^2$, $\forall x \in (0, 1)$

(C) $f(x) \cdot f^{-1}(x) = x^2$, has two roots in $(0, 1)$

(B) $f(x) \cdot f^{-1}(x) > x^2$, $\forall x \in (0, 1)$

(D) $f(x) \cdot f^{-1}(x) = x^2$, has exactly one root in $(0, 1)$.



$$\begin{aligned} m_{AC} &> m_{AB} \\ \Rightarrow \frac{x}{f(x)} &> \frac{f^{-1}(x)}{x} \end{aligned}$$

$$\Rightarrow f(x) \cdot f^{-1}(x) < x^2$$

(A) Correct

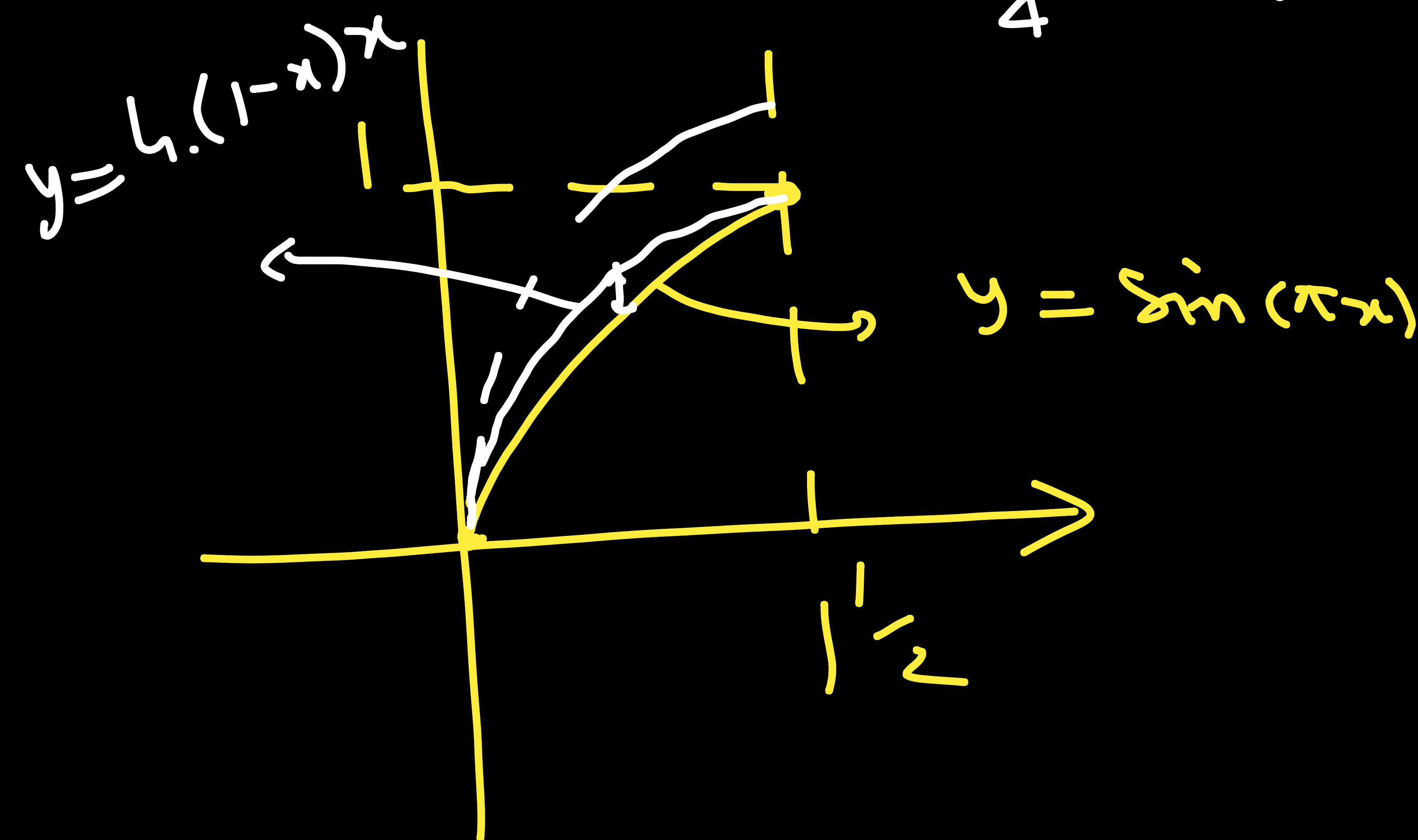
A

Ques If $\frac{\sin(\pi x)}{x(1-x)} \leq \lambda, \forall x \in (0,1)$ then find the least value of λ .

Soln.

$$\frac{\sin(\pi x)}{x(1-x)} \leq \lambda \Rightarrow \sin(\pi x) \leq \lambda(1-x)x$$

$$\frac{\lambda}{4} \geq 1 \Rightarrow \lambda \geq 4$$



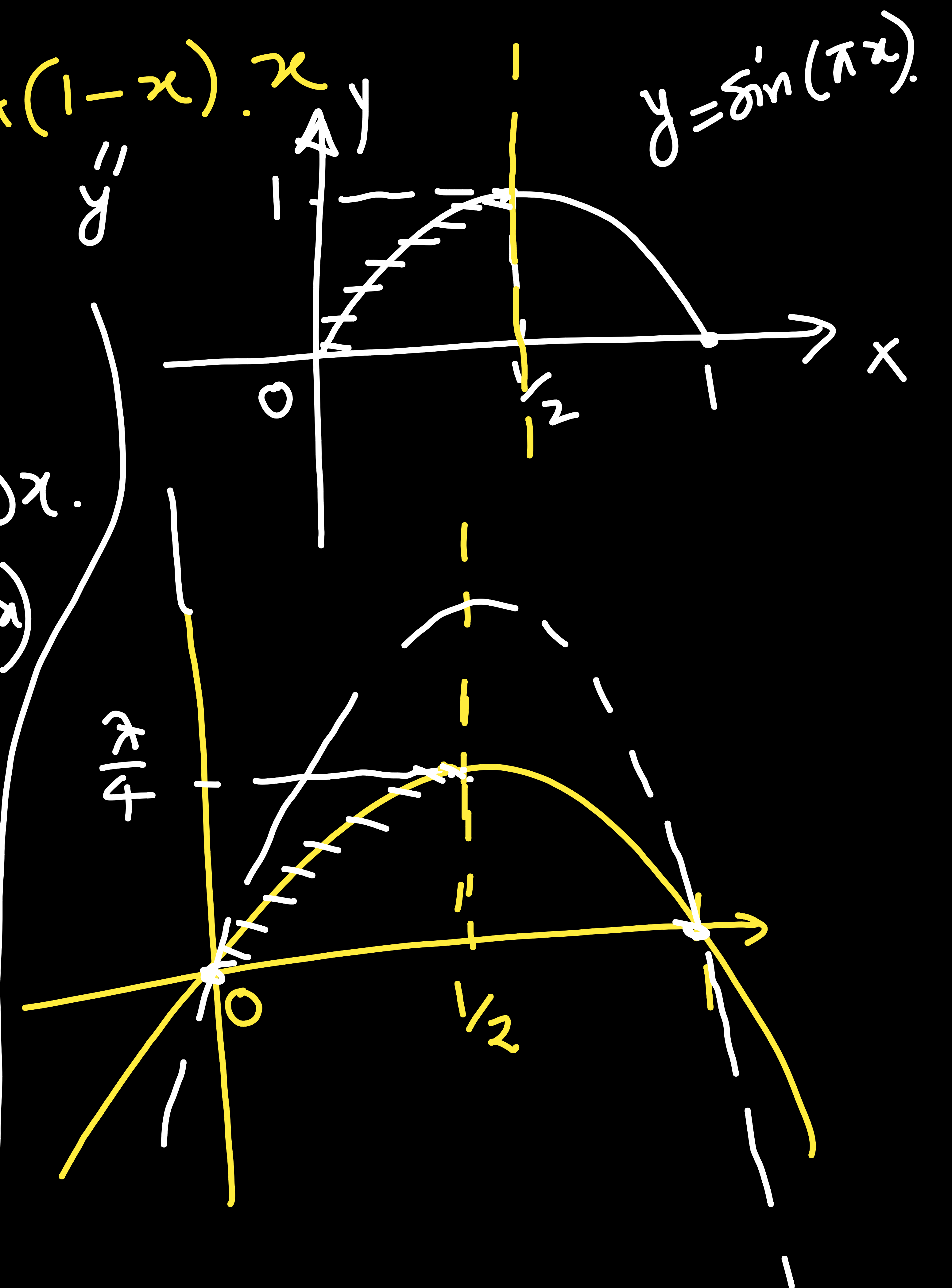
$$f(x) = \sin(\pi x) - \lambda(1-x)x$$

$$f'(x) = \pi \cos(\pi x) - \lambda(1-2x)$$

$$= \pi \cos(\pi x) + 2\lambda x - \lambda$$

↓ If $\lambda = 4$ then

$$f'(x) = \pi \cos(\pi x) + 8x - 4$$



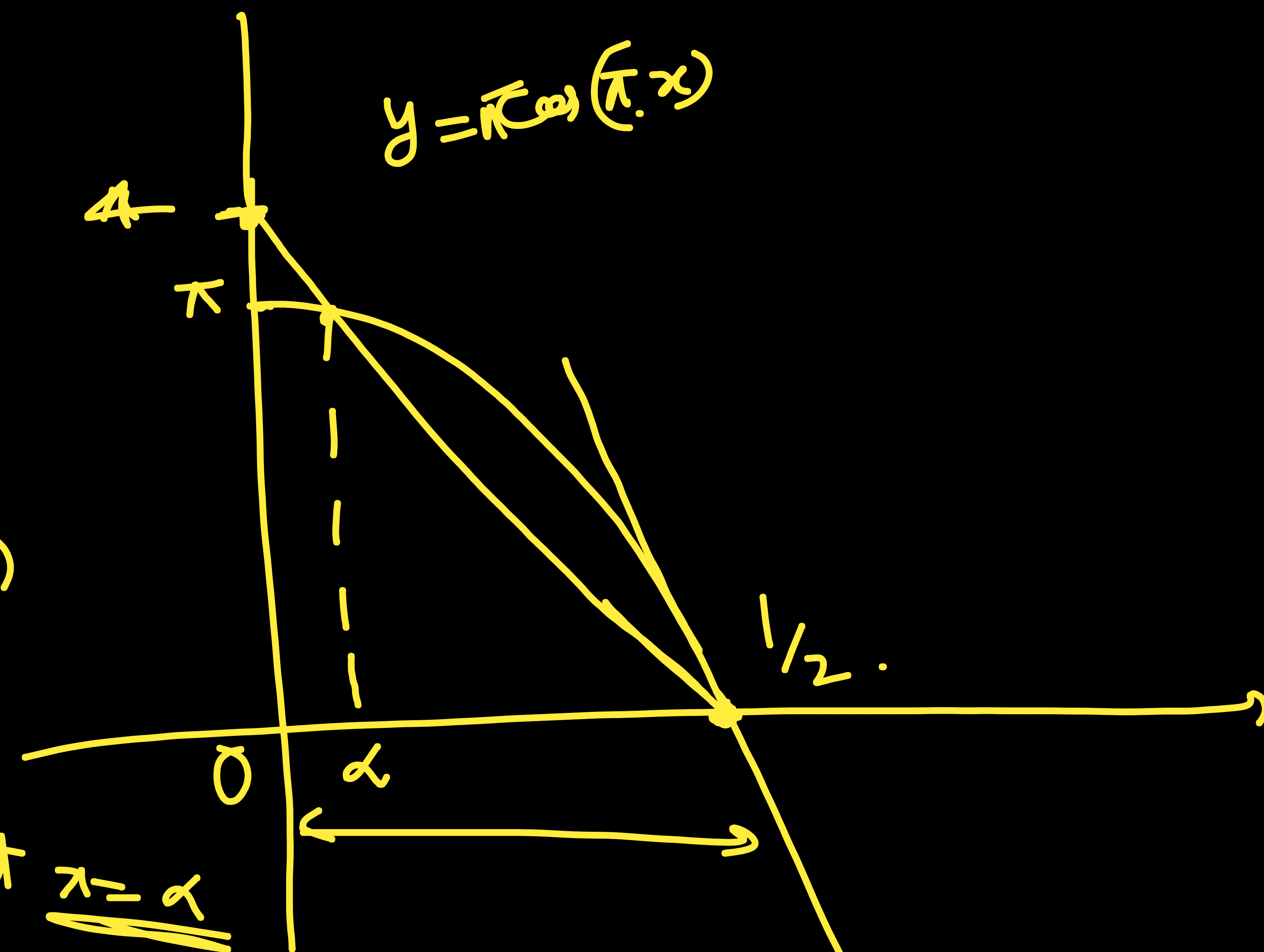
$$f'(x) = \pi \cos(\pi x) + \frac{8x - 4}{\underline{(x - 8x)}}$$

$$f'(0) = \pi - 4 < 0.$$

$$f'(\frac{1}{2}) = \pi \cos(\frac{\pi}{2}) + 4 - 4$$

$$= 0$$

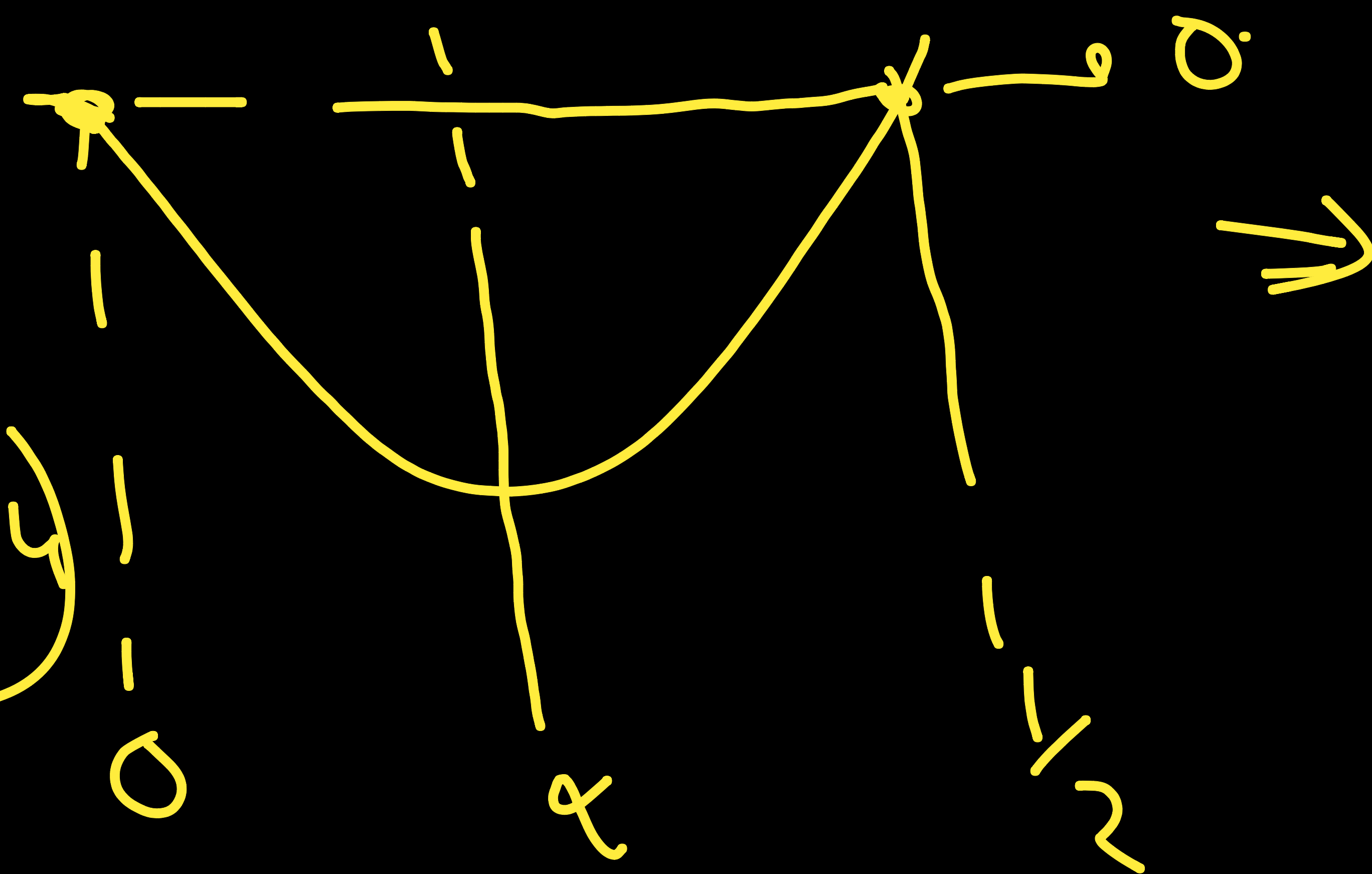
$\Rightarrow f'(x)$ has point of local min. at $x = \alpha$



$$\Rightarrow y = f(x)$$

$$f(0) = 0$$

$$f(\frac{1}{2}) = 0 \quad (\text{for } \lambda = 4)$$



$$f(x) \leq 0, \quad \forall x \in [0, \frac{1}{2}]$$

$$\Downarrow$$

$$\sin(\pi x) \leq 4x(1-x), \quad \forall x \in [0, \frac{1}{2}]$$

$$\Downarrow$$

$$\forall x \in [0, 1]$$

$$\boxed{\lambda_{\min} = 4}$$